Experiment 6: K-Means Clustering

Aim:

To perform K-Means clustering on the given dataset and group similar data points into clusters.

Algorithm:

1. Import the required libraries.
2. Load the dataset and select the required numerical features.
3. Create a K-Means clustering model with three clusters.
4. Fit the model to the dataset.
5. Obtain the cluster labels and centroids.
6. Visualize the clusters using a scatter plot.
7. Display the cluster centroids.

```python
X = data[['Hours_Studied', 'Marks']]

imputer = SimpleImputer(strategy='mean')
X = pd.DataFrame(imputer.fit_transform(X), columns=X.columns)

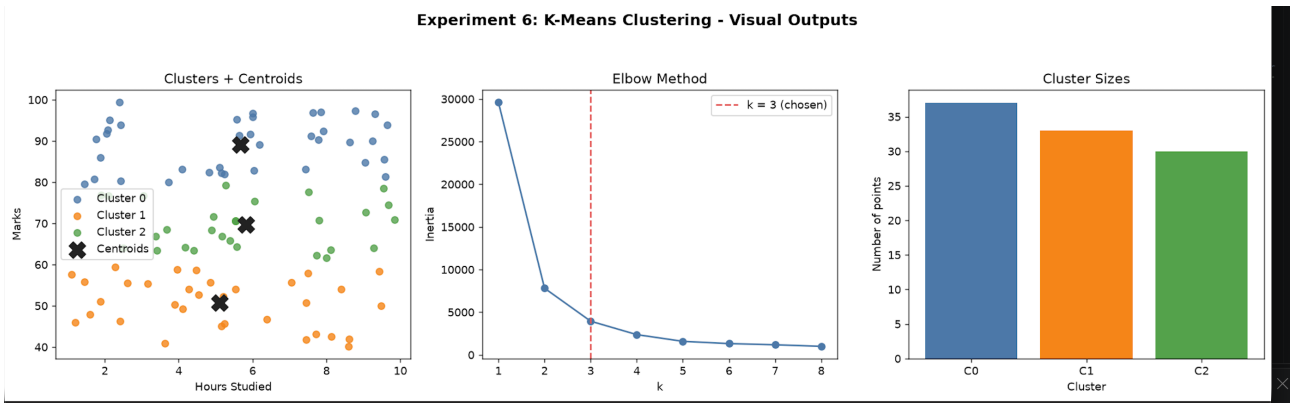
kmeans = KMeans(n_clusters=3, random_state=42, n_init=10)
kmeans.fit(X)

clusters = kmeans.labels_
data['Cluster'] = clusters

print(data.head())
print("\nCluster sizes:", np.bincount(clusters))
print("Inertia:", kmeans.inertia_)

# ---------- Hierarchical representation ----------
pipeline([
    ("1. Data", ["Hours_Studied", "Marks"]),
    ("2. Clean", ["Impute mean"]),
    ("3. K-Means", ["k = 3", "n_init = 10"]),
```

```
manasa@Manasass-MacBook FML Record 11259A119.py % python -u "/Users/manasa/codes/FML Record 11259A119.py/FML Record experiments/
FML Record 11259A119 Experiment 6.py"
Experiment 6: K-Means Clustering
├── Input Features
│   ├── Hours_Studied
│   ├── Marks
│   └── n = 100
├── Model
│   ├── KMeans(k=3)
│   ├── Inertia = 3946.0
│   └── random_state = 42
└── Clusters
    ├── Cluster 0: 37 points
    ├── Cluster 1: 33 points
    ├── Cluster 2: 30 points
    ├── Centroid 0 = (5.67, 89.09)
    ├── Centroid 1 = (5.11, 50.76)
    └── Centroid 2 = (5.82, 69.71)
```



Result:Thus, K-Means clustering was performed successfully.